

Devansh Khandar

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EDUCATION

Dwarkadas J. Sanghvi College of Engineering

Mumbai

B.Tech in Electronics and Telecommunication (Pursuing) | CGPA 8.45

B.Tech Honors in Artificial Intelligence and Machine Learning (Pursuing) | CGPA 8.2

EXPERIENCE

Electronics Trainee Engineer

June 2025 – October 2025

Prama Instruments Pvt. Ltd.

- Developed a real-time weather monitoring system integrating sensors for temperature, pressure, humidity, wind speed, rainfall, UV index, and ambient light.
- Built a custom 3D-printed wind-vane with Hall sensors for accurate wind-direction measurement.
- Improved system calibration and data reliability through modular sensor validation and testing.

Research & Development Engineer + Robotics Trainer

June 2024 – July 2024

RoboKart

- Conducted robotics training workshops across multiple colleges, teaching 300+ students.
- Delivered hands-on modules on sensors, automation, Arduino/WiFi systems, and introductory IoT.
- Led practical sessions on motion detection, home automation, servo control, and security systems.

PROJECTS

AI-Based Solar-Powered Farm Surveillance System

July 2025 – Present

- Built a Raspberry Pi + Google Coral edge-AI unit for real-time detection of bats, boars, monkeys, and deer using a custom MobileNet-SSD model (92% accuracy, <200 ms latency).
- Powered by solar panels with integrated BMS, battery storage, and DC-DC converters to ensure 72 hr autonomous operation.
- Automated ultrasonic, strobe-LED, and sprinkler deterrents via MQTT triggers; cut wildlife incursions by 35% in pilot tests.

Short-Horizon Return Prediction Challenge (i-RAGE)

May 2026

- Built a memory-optimized ML pipeline to predict short-horizon financial returns on 4 GB of high-frequency market data (2.4M+ samples).
- Engineered and benchmarked 5 prediction strategies including gradient boosting ensembles, ridge regression, and rank-based transformations.
- Identified a 44× prediction-scaling deficiency via statistical analysis, improving model output accuracy by 10–50×.
- Developed automated diagnostic and validation tooling to ensure data integrity and reproducibility across iterative experiments.

Autonomous Safari Navigation System

April 2026

- Developed an autonomous line-following vehicle using IR sensors and ESP32, capable of navigating a predefined safari track with high accuracy and real-time decision-making.
- Implemented real-time tracking and visualization system by integrating GPS and wireless communication to display vehicle trajectory, detected objects, and system status on a web-based dashboard.
- Designed an edge AI-based object detection pipeline using ESP32-CAM, combined with RFID-triggered event logging to generate automated reports of detected objects and mission completion.

ResQNet: IoT-Based Crisis Management System

March 2026

- Developed ResQNet, an ESP32-based IoT crisis response system with real-time detection, geolocation, and AI-assisted validation using a human-in-the-loop approach.
- Engineered a BLE mesh network for decentralized SOS relaying during outages, with offline data caching and cloud synchronization via gateway.
- Integrated multi-sensor monitoring (gas, water, rain, accelerometer) for real-time anomaly detection and emergency alerts.

Autonomous Drone Navigation & Charging System

March 2025

- Developed an autonomous drone capable of point-to-point navigation and auto-docking with its charging pod via sensor fusion and custom Flight Control System (FCS).
- Competed in a 48-hour Tech-a-Thon at Thadomal Shahani Engineering College, Mumbai; secured a Top-5 finish for building and demonstrating the end-to-end drone solution.
- Developed an Edge Impulse image model on ESP32-CAM for charging-pod detection (>95% on-device accuracy).

SKILLS

Programming Languages & Frameworks: C, C++, Java, JavaScript, Python, Linux (Ubuntu), MySQL

Softwares: AutoCAD, KiCAD, Arduino IDE, Altium Designer, Edge Impulse, Fusion 360, XCTU (XBee Telemetry), Tableau

Core ECE Tools: MATLAB, Simulink, LoRa, BLE, Xbee

POSITION OF RESPONSIBILITY

Internship & Placement Coordinator

June 2025 – Present

Dwarkadas J. Sanghvi College of Engineering

- Coordinating end-to-end campus placement and internship drives, serving as primary liaison between recruiters, HR teams, and students.
- Managing recruiter communications, scheduling, and preparation pipelines– demonstrating client relationship management and cross-functional coordination.

Electronics Senior (Avionics)

June 2024 – Present

DJS ARYA - The Official Canister Satellite Team of DJ Sanghvi

- Designed and routed multi-layer PCBs in Altium Designer for compact avionics payload integration.
- Developed flight software on Teensy 4.1 & ESP32, integrating GY-91 IMU, BME680, and encoder feedback systems.
- Implemented Kalman & Complementary filter fusion for high-frequency telemetry over XBEE - field tested up to 1 km range.
- Built closed-loop PID control with SimpleFOC for BLDC motor actuation during mechanical deployment sequences.

Technical Dept. Member

August 2024 – August 2025

DJSCE IETE-ISF

- En-Core 1.0 (Sep 24) - 8-hr analog hackathon (PWM DC-motor speed-control challenge).
- Unplugged 2.0 (Mar '25) - 24-hr, 3-round hardware hackathon (PPT, KiCAD PCB, prototyping).
- Mentored teams on PCB fabrication, KiCAD & Fusion 360; managed event logistics.

AWARDS & ACHIEVEMENTS

- Won the **Mark Walker Best Ground Station Award 2026** at the **International CanSat Competition** for developing a reliable custom ground station with real-time data visualization, antenna tracking, and telemetry management.
- International CanSat Competition - Ranked 1st in India, 4th in Asia and 10th worldwide (2026) among 400+ teams organized by AAS in collaboration with NASA
- International CanSat Competition - Ranked 1st in India and 5th worldwide (2025) and 1st in Asia and 12th worldwide (2024) among 400+ teams organized by AAS in collaboration with NASA
- IN-SPACe CanSat India Student Competition (2024-25) - Secured All India Rank 1 among 70+ teams organized by ASI in collaboration with ISRO, awarded for flawless subsystem integration and payload performance.
- Medal for Excellence in Internship by GUJCOST, awarded by Dr. Rajesh Modi for outstanding performance as a robotics trainer.
- Published aeroponic IoT research in DJS STRIKE 2024-25 conference proceedings (ISBN: 978-93-341-2968-7).